

GENERAL INFORMATION

DIAGNOSTIC TROUBLE CODE INDEX - DTC:
TRANSFER CASE CONTROL MODULE (TCCM) [G1817185]

DESCRIPTION AND OPERATION

TRANSFER CASE CONTROL MODULE (TCCM)

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault, and may also cause additional faults in the vehicle being tested and/or the donor vehicle.

NOTES:

- Guided Diagnostics must be followed and the repair advised by the process completed. Failure to do so may result in the rejection of any warranty claim made.
- If a control module or a component is at fault and the vehicle remains under manufacturer warranty, refer to the Warranty Policy and Procedures manual, or determine if any prior approval program is in operation, prior to the installation of a new module/component.
- Generic scan tools may not read the codes listed, or may read only 5-digit codes. Match the 5 digits from the scan tool to the first 5 digits of the 7-digit code listed to identify the fault (the last 2 digits give extra information read by the manufacturer-approved diagnostic system).
- When performing voltage or resistance tests, always use a digital multimeter accurate to three decimal places, and with an up-to-date calibration certificate. When testing resistance always take the resistance of the digital multimeter leads into account.
- Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.
- Inspect connectors for signs of water ingress, and pins for damage and/or corrosion.
- If Diagnostic Trouble Code(s) (DTC)s are recorded and, after performing the pinpoint tests, a fault is not present, an intermittent concern may be the cause. Always check for loose connections and corroded terminals.
- Check Jaguar Land Rover (JLR) claims submission system for open campaigns. Refer to the corresponding bulletins and SSMS which may be valid for the specific customer complaint and complete the recommendations as required.

The table below lists all Diagnostic Trouble Codes (DTCs) that could be set in the Transfer Case Control Module (TCCM). For additional diagnosis and testing information, refer to the relevant Diagnosis and Testing section in the workshop manual. For additional information, refer to: [Transfer Case](#) (307-07 Transfer Case, Diagnosis and Testing).

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
B1142-62	Ignition Status 1 - Signal compare failure	NOTE: Circuit reference - IGN - Ignition signal circuit short circuit to ground	Refer to the electrical circuit diagrams and check the ignition signal circuit for short circuit to ground. Repair the wiring harness as necessary
P080A-56	Clutch Position Not Learned - Invalid/incomplete configuration	- Transfer case clutch not calibrated - Transfer case control module internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module
P0810-04	Clutch Position Control Error - System internal failures	- Transfer case clutch not calibrated - Transfer case control module internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module
P0810-07	Clutch Position Control Error - Mechanical failures	- Transfer case classification data missing - Transfer case clutch not calibrated - Transfer case control module internal failure - Transfer case internal failure	NOTE: The classification data is displayed on a label on the transfer case. - Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Transfer Case Classification - Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case
P0810-56	Clutch Position Control Error - Invalid/incomplete configuration	- Transfer case clutch not calibrated - Transfer case control module internal failure - Transfer case internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case
P0810-78	Clutch Position Control Error - Alignment or adjustment incorrect	- Transfer case clutch not calibrated - Transfer case control module internal failure - Transfer case internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
P170E-92	Clutch Delivered Torque Performance - Performance or incorrect operation	- Transfer case clutch not calibrated - Transfer case control module internal failure - Transfer case internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Calibrate Clutch - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case
P173A-64	Clutch Actuator Position Sensor Circuit Range/Performance - Signal plausibility failure	- Transfer case control module internal failure - Transfer case internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case
P175B-00	Transfer Case Clutch Deteriorated - No sub type information	Transfer case internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case
P175C-00	Transfer Case Fluid Deteriorated - No sub type information	Transfer case fluid deteriorated	Refer to the relevant section of the workshop manual and drain and refill the transfer case fluid
P215A-62	Vehicle Speed / Wheel Speed Correlation - Signal compare failure	- One or more wheel/tire sizes incorrect - Missing/invalid data from the anti-lock brake system control module	- Check that the wheels and tires comply with the manufacturer's specification for the vehicle. Rectify as necessary - Using the Jaguar Land Rover approved diagnostic equipment, check the anti-lock brake system control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
P2787-98	Clutch Temperature Too High - Component or system over temperature	Transfer case clutch temperature above maximum threshold	NOTES: - This <u>DTC</u> is for event information only and does not indicate a fault. - This <u>DTC</u> may be induced by prolonged driving on a poor traction surface. - Allow the system temperature to return to normal. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest
U0001-81	High Speed Controller Area Network (CAN) Communication Bus - Invalid serial data received	Invalid data received from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-82	High Speed <u>CAN</u> Communication Bus - Alive/sequence counter incorrect / not updated	Invalid data received from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U0001-83	High Speed <u>CAN</u> Communication Bus - Value of signal protection calculation incorrect	Invalid data received from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the invalid data source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-87	High Speed <u>CAN</u> Communication Bus - Missing message	Missing message from another control module through the high speed <u>CAN</u> bus (powertrain)	Using the Jaguar Land Rover approved diagnostic equipment, check the snapshot data to determine the missing message source control module. Check the relevant control module for related <u>DTCs</u> and refer to the relevant <u>DTC</u> index
U0001-88	High Speed <u>CAN</u> Communication Bus - Bus off	High speed <u>CAN</u> bus (powertrain) circuit short circuit to ground, short circuit to power, open circuit, high resistance	Using the Jaguar Land Rover approved diagnostic equipment, perform a <u>CAN</u> network integrity test. Refer to the electrical circuit diagrams and check the high speed <u>CAN</u> bus (powertrain) circuit for short circuit to ground, short circuit to power, open circuit, high resistance. Repair the wiring harness as necessary
U0300-00	Internal Control Module Software Incompatibility - No sub type information	- Transfer case control module is not configured correctly - Incorrect transfer case control module installed	- Using the Jaguar Land Rover approved diagnostic equipment, re-configure the transfer case control module with the latest level software - Check the part number of the transfer case control module. Install a new transfer case control module as necessary
U2012-4A	Car Configuration Parameter(s) - Incorrect component installed	Car configuration file mismatch with vehicle specification	NOTE: After updating the car configuration file, set the ignition to on and wait 30 seconds before clearing the <u>DTCs</u>. Using the Jaguar Land Rover approved diagnostic equipment, check and update the car configuration file as necessary. Clear the <u>DTCs</u> and retest
U2100-54	Initial Configuration Not Complete - Missing calibration	Transfer case classification data missing	NOTE: The classification data is displayed on a label on the transfer case. Using the Jaguar Land Rover approved diagnostic equipment, perform routine - Transfer Case Classification
U3000-47	Control Module - Watchdog/safety microcontroller failure	NOTES: - Circuit reference - VBATT1 - - Circuit reference - VBATT2 - - Circuit reference - GND - - Transfer case control module power or ground circuit open circuit, high resistance - Transfer case control module internal failure	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3000-49	Control Module - Internal electronic failure	Transfer case control module internal failure	Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest. If the fault persists, install a new transfer case control module
U3000-4B	Control Module - Over temperature	Transfer case control module internal temperature above maximum threshold	NOTES: - This <u>DTC</u> is for event information only and does not indicate a fault. - This <u>DTC</u> may be induced by prolonged driving on a poor traction surface. - Allow the system temperature to return to normal. Using the Jaguar Land Rover approved diagnostic equipment, clear the <u>DTCs</u> and retest
U3006-13	Control Module Input Power A - Circuit open	NOTES: - Circuit reference - VBATT1 - - Circuit reference - GND - - Transfer case control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3006-16	Control Module Input Power A - Circuit voltage below threshold	NOTES: - Circuit reference - VBATT1 - - Circuit reference - GND - - Transfer case control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3006-17	Control Module Input Power A - Circuit voltage above threshold	Battery/charging system fault	Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD018). Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3006-62	Control Module Input Power A - Signal compare failure	NOTES: - Circuit reference - VBATT1 - - Circuit reference - GND - - Mismatch between the voltage at the transfer case control module and the voltage value broadcast on the <u>CAN</u> bus	Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD018) - and compare it to battery voltage. Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary

DTC	DESCRIPTION	POSSIBLE CAUSES	ACTION
U3007-13	Control Module Input Power B - Circuit open	NOTES: - Circuit reference - VBATT2 - - Circuit reference - GND - - Transfer case control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD019). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3007-16	Control Module Input Power B - Circuit voltage below threshold	NOTES: - Circuit reference - VBATT2 - - Circuit reference - GND - - Transfer case control module power or ground circuit open circuit, high resistance - Battery/charging system fault	- Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD019). Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary - Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3007-17	Control Module Input Power B - Circuit voltage above threshold	Battery/charging system fault	Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD019). Refer to the relevant section of the workshop manual and check the battery and charging system. Rectify as necessary
U3007-62	Control Module Input Power B - Signal compare failure	NOTES: - Circuit reference - VBATT2 - - Circuit reference - GND - - Mismatch between the voltage at the transfer case control module and the voltage value broadcast on the <u>CAN</u> bus	Using the Jaguar Land Rover approved diagnostic equipment, check datalogger signal - Main ECU Supply Voltage (0xD019) - and compare it to battery voltage. Refer to the electrical circuit diagrams and check the transfer case control module power and ground circuits for open circuit, high resistance. Repair the wiring harness as necessary